

Qualification Pack



Technical Helper - Thermal Power Generation

QP Code: PSS/Q1701

Version: 1.0

NSQF Level: 2

Power Sector Skill Council || Plot No. 4, Institutional Area, CBIP Building, Malcha Marg, Chanakyapuri
New Delhi-110021 || email: akbhagatin@yahoo.co.in



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PSS/Q1701: Technical Helper - Thermal Power Generation

Brief Job Description

The individual in this job helps technical supervisor/operator/maintenance team member by performing supporting activities such as material handling, supplying or holding tools, lifting and carrying objects, oil & clean machinery and equipment etc.

Personal Attributes

The individual must be physically and mentally able to perform essential functions of the job safely. The candidate should be able to climb ladders, scaffolds and poles of various heights. The candidate should be able to read, listen carefully to and understand instructions and warnings.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1331: Apply basic health and safety practices for power related work](#)
2. [PSS/N1336: Working effectively with others](#)
3. [PSS/N1701: Perform basic tasks for repair and maintenance of power generation system and components \(Electrical\)](#)
4. [PSS/N1702: Perform basic tasks for repair and maintenance of equipment and components \(Mechanical\)](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Generation
Occupation	Operation and Maintenance, Erection, Installation and Commissioning - Generation
Country	India
NSQF Level	2
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7413.9900



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Minimum Educational Qualification & Experience	8th Class
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	11/02/2019
Next Review Date	10/02/2023
NSQC Approval Date	22/08/2019
Version	1.0
Reference code on NQR	2019/POW/PSSC/03473
NQR Version	1.0



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PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Scope

This unit/task covers the following:

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** • use protective clothing/equipment for specific tasks and work conditions
Protective Clothing
• leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visor
equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace
Hazards- electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions
- PC3.** follow procedures for safe working with electrical equipment
Safe Working Procedures: tag out/lock out, ptw (permit to work)
- PC4.** follow warning signs while working with electrical systems
Warning Signs: danger, high voltage area, out of service, etc
- PC5.** use standard safe working practices when working at heights, confined areas and trenches
- PC6.** test any electrical equipment and system using insulated testing devices before touching them
- PC7.** ensure positive isolation of electrical equipment & system as per the given standards
- PC8.** identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installed
Parameters: temperature, pressure, flow and current

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- PC9.** follow safe working practices while working in hazardous conditions to ensure the safety of self and others
Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.
- PC10.** use various methods of accident prevention in the work environment
Methods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors
- PC11.** identify and retrieve general health and safety equipment in the workplace
General Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.
- PC12.** set up and use scaffolds, elevated platforms and ladders safely
Set up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc
- PC13.** inspect scaffolds, elevated platforms and ladders for faults and rectify them
Faults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.
- PC14.** lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa
- PC15.** inspect power plant and its equipment routinely for any signs of oil, water and steam leakage
- PC16.** store flammable materials and machine lubricating oil safely and correctly
- PC17.** check that the emission and pollution control devices are working properly in line with environmental policy standards
- PC18.** use good housekeeping practices at all times
Good Housekeeping Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces
- PC19.** identify common hazard signs displayed in various areas
Various Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.
- PC20.** inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

- PC21.** use various appropriate fire extinguishers for different types of fires correctly
Types of Fires: class a, b, c, d and e
- PC22.** use appropriate rescue techniques during fire hazard
- PC23.** maintain good housekeeping in order to prevent fire hazards
- PC24.** use a fire extinguisher correctly in case of fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC25.** administer appropriate first aid to the victims for the relevant injuries
Injuries: bleeding, burns, choking, electric shock, poisoning etc.
- PC26.** act promptly and appropriately to an accident situation or medical emergency in real or simulated environments



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- PC27.** perform and organise loss minimization or rescue activity during an accident in real or simulated environments
- PC28.** administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services
- PC29.** demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergencies
Emergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work
- PC30.** free an affected person from electrocution safely as per the recommended procedure
- PC31.** move injured people and others to a safe place during emergencies
- PC32.** write an incident report or dictate a report to another person, and send report to the authorized person
Incident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. names (and job titles if applicable), and where to find, all the people responsible for health and safety in a workplace
2. names and location of documents that refer to health and safety in the workplace
3. names and location of people responsible for health and safety in the workplace
4. documents that refer to health and safety in the workplace
documents: fire notices, accident reports, safety instructions for equipment and procedures, company notices and documents, legal documents (e.g. government notices)
5. meaning of hazards and risks
6. health and safety hazards commonly present in the work environment and related precautions
7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
8. possible causes of risk and accident
possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness); not taking safety precautions
9. methods of accident prevention
10. safe working practices when working with tools and machines
11. safe working practices while working at various hazardous sites
12. where to find all the general health and safety equipment in the workplace
13. various dangers associated with the use of electrical equipment
14. positive isolation of electrical equipment and system
15. safe handling and disposal of hazardous power plant wastes
16. use of emission and pollution control devices and measures taken to control pollution
17. various safety procedures and equipment used to work at heights, trenches and confined places
18. safe working practices specific to working with electrical equipment and system e.g. lock out/tag out, ptw, etc.



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19. preventative and remedial actions to be taken in the case of exposure to toxic materialstoxic materials: solvents, flux, lead exposure: ingested, contact with skin, inhaledpreventative action: ventilation, masks, protective clothing/ equipment); remedial action: immediate first aid, report to supervisor
20. importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
21. precautionary measures taken to prevent fire accident
22. various causes of firecauses of fires: heating of metal; spontaneous ignition; sparking; electrical heating; loose fires (smoking, welding, etc.); chemical fires; etc.
23. correct technique(s) of using a fire extinguisher
24. different methods of extinguishing fire
25. different materials used for extinguishing firematerials: sand, water, foam, co2, dry powder
26. emergency rescue techniques applied during a fire hazard
27. various types of safety signs and what they mean
28. appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
29. rescue techniques applied during fire hazards
30. good housekeeping in order to prevent fire hazards
31. correct use of a fire extinguisher
32. how to free a person from electrocution
33. basic techniques of bandaging
34. perform artificial respiration and the cpr process

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write an accident/ incident report in local language and/or english
2. read and comprehend basic content to read labels, charts, signages
3. read and comprehend basic english to read manuals of operation
4. read an accident/ incident report in local language or english
5. question coworkers appropriately in order to clarify instructions and other issues
6. give clear instructions to coworkers and subordinates
7. remain congenial while discussing and debating issues with co-workers
8. follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
9. ask for, provide and receive required assistance where possible to ensure achievement of work-related objectives
10. follow organizational rule-based decision-making process
11. take decisions with systematic course of actions and/or response
12. plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
13. build customer relationships and use customer centric approach



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14. think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
15. seek appropriate assistance from other sources to resolve problems
16. report problems that cannot be resolved to the appropriate authority
17. identify cause and effect relations in own area of work
18. work systematically and logically to resolve the issues, identify the cause and anticipate unexpected results
19. critically evaluate the organizational processes to improve them

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	19	43	-	-
PC1. - use protective clothing/equipment for specific tasks and work conditions - Protective Clothing - leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visor - equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator	1	2	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace Hazards-electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions	1	2	-	-
PC3. follow procedures for safe working with electrical equipment Safe Working Procedures: tag out/lock out, ptw (permit to work)	1	2	-	-
PC4. follow warning signs while working with electrical systems Warning Signs: danger, high voltage area, out of service, etc	1	2	-	-
PC5. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC6. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC7. ensure positive isolation of electrical equipment & system as per the given standards	1	1	-	-
PC8. identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installed Parameters: temperature, pressure, flow and current	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. follow safe working practices while working in hazardous conditions to ensure the safety of self and others Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.	1	2	-	-
PC10. use various methods of accident prevention in the work environment Methods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors	1	2	-	-
PC11. identify and retrieve general health and safety equipment in the workplace General Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.	1	2	-	-
PC12. set up and use scaffolds, elevated platforms and ladders safely Set up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc	1	3	-	-
PC13. inspect scaffolds, elevated platforms and ladders for faults and rectify them Faults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/unfixed nuts or bolts, etc.	1	3	-	-
PC14. lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa	-	3	-	-
PC15. inspect power plant and its equipment routinely for any signs of oil, water and steam leakage	1	3	-	-
PC16. store flammable materials and machine lubricating oil safely and correctly	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	2	-	-
PC18. use good housekeeping practices at all times Good Housekeeping Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces	1	2	-	-
PC19. identify common hazard signs displayed in various areas Various Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.	1	2	-	-
PC20. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	2	-	-
<i>Follow fire safety practices</i>	3	8	-	-
PC21. use various appropriate fire extinguishers for different types of fires correctly Types of Fires: class a, b, c, d and e	1	2	-	-
PC22. use appropriate rescue techniques during fire hazard	1	2	-	-
PC23. maintain good housekeeping in order to prevent fire hazards	1	2	-	-
PC24. use a fire extinguisher correctly in case of fire	-	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	19	-	-
PC25. administer appropriate first aid to the victims for the relevant injuries Injuries: bleeding, burns, choking, electric shock, poisoning etc.	1	2	-	-
PC26. act promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC27. perform and organise loss minimization or rescue activity during an accident in real or simulated environments	1	4	-	-
PC28. administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergencies Emergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC30. free an affected person from electrocution safely as per the recommended procedure	1	2	-	-
PC31. move injured people and others to a safe place during emergencies	-	3	-	-
PC32. write an incident report or dictate a report to another person, and send report to the authorized person Incident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified	2	2	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Distribution, Transmission, Generation, Distribution Downstream
Occupation	Operation & Maintenance - Generation Supervision - Distribution Erection, Installation and Commissioning - Distribution Surveying - Transmission Operation & Maintenance - Transmission
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	11/02/2019
Next Review Date	11/02/2023
NSQC Clearance Date	22/08/2019



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PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and co-ordinate with in the organization
- KU6.** importance of effective communication in the workplace
- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication



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- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge
- GS2.** note down observations (if any) related to the operation/maintenance
- GS3.** read and interpret the process required for different types of manuals
- GS4.** read and interpret the flowchart of all parts of an assembly
- GS5.** read manuals and documents to understand the product-details & how they can be used
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor
- GS7.** effectively communicate with the team members
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor
- GS9.** communicate clearly with the colleague on the issues faced during query/fault
- GS10.** follow colleague/contractor rule-based decision making process
- GS11.** take decisions with systematic course of actions and/or response
- GS12.** planning and organization of tasks to meet deadlines
- GS13.** build customer relationships and use customer centric approach
- GS14.** seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
- GS15.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
- GS16.** critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>working with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023



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PSS/N1701: Perform basic tasks for repair and maintenance of power generation system and components (Electrical)

Description

This unit is about the competencies required by the Technical Helper to perform basic tasks to assist plant operator/supervisor/maintenance team member in plant operations, inspection, repair and maintenance of electrical equipment.

Scope

This unit/ task covers the following:

- Handling of tools and equipment
- Carry out basic tasks related to repair and maintenance activities

Elements and Performance Criteria

Handle tools and equipment

To be competent, the user/individual on the job must be able to:

- PC1.** identify the tools and equipment required for performing various repair and maintenance activities
- PC2.** obtain the required tools and equipment following organizational procedure
- PC3.** ensure the tools and equipment are oiled, cleaned, well maintained, calibrated and ready for use

Carry out basic tasks related to electrical repair and maintenance

To be competent, the user/individual on the job must be able to:

- PC4.** prepare and maintain the work area as per instructions, procedure or operation specification
- PC5.** perform preparatory and post activity tasks when supporting in installation, commissioning, operations and maintenance activities
- PC6.** perform various activities to assist supervisor/operator/maintenance team member in carrying out Operation and Maintenance (O and M) tasks Activities: Holding, handing over components, measuring, greasing, transportation, loading and unloading, etc.
- PC7.** perform cable pulling, cable jointing and clamping
- PC8.** measure, cut, and bend wire and conduit using measuring instruments and hand tools
- PC9.** tighten loose connections and repair broken insulation using hand tools and as per the guidance provided
- PC10.** drill holes and pull or push wiring through openings
- PC11.** perform various activities for preparation and adjustment of tools and equipment for use Preparation and adjustment activities: Raise, lower, or position tools, equipment and materials, using hoist, hand line, or block and tackle
- PC12.** paint a variety of objects related to electrical functions
- PC13.** perform various repair and maintenance activities using tools and equipment Repair and maintenance activities: Hammering, tightening, digging, cleaning measuring, pipe cutting etc.

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- PC14.** detect various signs of fault in the equipment Signs of fault: Vibrations, excessive sound, oil, air, gas, water and steam leakage or any similar abnormality
- PC15.** follow necessary precautions while working in areas surrounded by chemicals
- PC16.** leave the work area in a safe and tidy condition on completion of the repair and maintenance activities
- PC17.** refer unresolved job-related problems to the appropriate personnel for support
- PC18.** monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the
2. organization relevant to own employment and performance conditions
3. relevant health and safety requirements applicable in the work place
4. own job role and responsibilities and sources for information pertaining to
5. employment terms, entitlements, job role and responsibilities
6. importance of working in a clean environment and safe practices and procedures
7. relevant people and their responsibilities within the work area
8. escalation matrix and procedures for reporting work and employment related issues
9. various types of chemicals and hazards of working with them
10. common electricity terminology Terminology: Current, voltage, resistance, kilowatt (kw), kilowatt hour(kwh)
11. tools and equipment used in testing, repair and maintenance activities Tools: Keys, Multimeter, Clampmeter, Plier, Screwdriver, Wrench set, Hammer, Drilling machine, Hacksaw / cutting tools, Measuring tape, Pulleys (Force Pulley with sling), Tommy bar, Crimping machine, Round / flat file, Earth rod (discharge rod), leakage current monitoring kit
12. correct use of various tools Tools: spanners, hammer, chisels, gauge and other basic tools required in a power plant
13. specific health and safety precautions which must be taken when carrying out repair and maintenance work Health and safety precautions: Avoidance of loose clothes, key chain, shoes with projecting nails and other types of metal parts; not to start work unless circuit is in off and earther condition
14. importance of obtaining permit to work before initiating maintenance or repair activity or equipment should be first made off and take permit to work before taking the work in hand
15. need for an authorized permit to work (PTW)
16. functioning of ammeters, volt meters, meggars, multi-meters
17. difference between rotary parts and static parts
18. hazards associated with working in a high voltage power plant and key mitigation measures Hazards: live wires, faulty insulation, voltage surges, faulty and damaged equipment and components, unsecure cables, unstable ladders, insects and reptiles, and scaffolding, etc.
19. importance of ensuring that tools and equipment are suitable, well maintained, calibrated and operating effectively



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20. importance of following good housekeeping and fire prevention procedures
21. importance of following job instructions
22. importance of leaving the work area and equipment in a safe and clean condition on completion of the repair and maintenance activities
23. importance of reporting problems in a timely manner
24. PPE (Personal Protective Equipment) and clothing that must be worn during the inspection, repair and maintenance activity and from where can it be obtained PPE: Safety helmet, safety glove, safety shoe, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), safety rope, etc.
25. standard procedure on how to deal with electric shocks and electrocutions
26. how to rescue a person from electrocution and minimize harm

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write notes to record information to be used at work including instructions by the supervisor
2. write observations related to the process, if any
3. read dimensions from drawings and diagrams correctly
4. read instructions posted on the equipment/walls/ sign boards etc.
5. communicate clearly with team and other staff members
6. discuss task lists, schedules and activities with the team members
7. make appropriate work-related judgments based on laid down policies and procedures
8. plan and organize tasks to meet quantity and quality specifications
9. identify impact of own work and actions on end customer
10. identify changes in conditions, operations, and the environment to anticipate and address problems

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Handle tools and equipment</i>	3	10	-	-
PC1. identify the tools and equipment required for performing various repair and maintenance activities	1	3	-	-
PC2. obtain the required tools and equipment following organizational procedure	1	4	-	-
PC3. ensure the tools and equipment are oiled, cleaned, well maintained, calibrated and ready for use	1	3	-	-
<i>Carry out basic tasks related to electrical repair and maintenance</i>	27	60	-	-
PC4. prepare and maintain the work area as per instructions, procedure or operation specification	2	4	-	-
PC5. perform preparatory and post activity tasks when supporting in installation, commissioning, operations and maintenance activities	2	4	-	-
PC6. perform various activities to assist supervisor/operator/maintenance team member in carrying out Operation and Maintenance (O and M) tasks Activities: Holding, handing over components, measuring, greasing, transportation, loading and unloading, etc.	2	4	-	-
PC7. perform cable pulling, cable jointing and clamping	2	4	-	-
PC8. measure, cut, and bend wire and conduit using measuring instruments and hand tools	2	4	-	-
PC9. tighten loose connections and repair broken insulation using hand tools and as per the guidance provided	2	4	-	-
PC10. drill holes and pull or push wiring through openings	1	4	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. perform various activities for preparation and adjustment of tools and equipment for use Preparation and adjustment activities: Raise, lower, or position tools, equipment and materials, using hoist, hand line, or block and tackle	2	5	-	-
PC12. paint a variety of objects related to electrical functions	1	4	-	-
PC13. perform various repair and maintenance activities using tools and equipment Repair and maintenance activities: Hammering, tightening, digging, cleaning measuring, pipe cutting etc.	2	4	-	-
PC14. detect various signs of fault in the equipment Signs of fault: Vibrations, excessive sound, oil, air, gas, water and steam leakage or any similar abnormality	2	4	-	-
PC15. follow necessary precautions while working in areas surrounded by chemicals	2	4	-	-
PC16. leave the work area in a safe and tidy condition on completion of the repair and maintenance activities	1	3	-	-
PC17. refer unresolved job-related problems to the appropriate personnel for support	2	4	-	-
PC18. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem	2	4	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1701
NOS Name	Perform basic tasks for repair and maintenance of power generation system and components (Electrical)
Sector	Power
Sub-Sector	Generation
Occupation	Operation & Maintenance; Erection, Installation and Commissioning Generation
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	11/02/2019
Next Review Date	10/02/2023
NSQC Clearance Date	22/08/2019



Qualification Pack

PSS/N1702: Perform basic tasks for repair and maintenance of equipment and components (Mechanical)

Description

This unit is about the competencies required for technical helpers (mechanical) involved in performing basic tasks to assist mechanics or mechanical operators in operations, repair and maintenance of mechanical equipment

Scope

This unit/ task covers the following:

- Perform basic tasks in maintenance and repair activities

Elements and Performance Criteria

Carry out basic tasks related to mechanical maintenance and repair

To be competent, the user/individual on the job must be able to:

- PC1.** prepare and maintain the work area as per procedures or operational specification
- PC2.** identify the tools and equipment required for operation and maintenance of mechanical system in thermal power plant
- PC3.** obtain the tools and equipment identified following the organizational procedure
- PC4.** perform various activities to help the mechanic or engineer Activities: holding, handing over components, measuring, loading and unloading, etc. while assisting mechanics or mechanical operators in carrying out replacement of components such as radiators, hoses, fan belts, batteries, spark plugs, ignition cables, and other parts
- PC5.** assemble and disassemble valves, pumps, motors and other components to ensure proper functioning of the overall system
- PC6.** perform lubrication of various tools and related activities in accordance with the established procedures and charts Activities: Check and change oil, oil filters and air filters; fuel and service buses and other vehicles; replace hoses, belts, water pumps, batteries and similar equipment and parts
- PC7.** perform a regular condition check of various components and equipment Components and equipment: Fan belts, hoses, wiper blades and safety equipment; radiators and batteries
- PC8.** prepare and adjust various tools Prepare and adjust: Raise, lower, or position tools, equipment and materials using hoist or hand line
- PC9.** paint various tools related to electrical functions, as instructed
- PC10.** detect various signs of fault in the equipment Signs of fault: Vibrations, excessive sound, oil, air, gas, water and steam leakage or any similar abnormality
- PC11.** leave the work area in a safe and tidy condition on completion of the repair and maintenance activities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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1. relevant legislation, standards, policies, and procedures followed in the company
2. relevant health and safety requirements applicable in the work place
3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
4. reporting structure, inter-dependent functions, lines and procedures in the work area
5. escalation matrix and procedures for reporting work and employment related issues
6. use and functions of the tools and equipment commonly found in machine, sheet metal, structural steel, automotive repair, and other mechanical shops
7. various types of chemicals and hazards involved in working with them
8. use and correct operation of various tools Tools: Spanners, hammers, chisels, gauge etc.
9. common electricity terminology Terminology: Current, voltage, resistance, kilowatt (kw), kilowatt hour(kwh)
10. tools and equipment used in testing, repair and maintenance Tools: Keys, Multimeter, Clampmeter, Plier, Screwdriver, Wrench set, Hammer, Drilling machine, Hacksaw / cutting tools, Measuring tape, Pulleys (Force Pulley with sling), Tommy bar, Crimping machine, Round / flat file, Earth rod (discharge rod), leakage current monitoring kit
11. importance of taking permit to work before initiating any maintenance and repair activities
12. specific health and safety precautions which must be taken when carrying out repair and maintenance work Health and safety precautions: avoid loose clothes, key chain or watch chains, shoes with projecting nails, other types of metal parts; not to start work unless circuit is in off and earther condition
13. hazards associated with working in a power plant and key mitigation measures Hazards: Live wires, faulty insulation, voltage surges, faulty and damaged equipment and components, unsecure cables, unstable ladders, insects and reptiles, and scaffolding, etc.
14. how to operate power hacksaws, emery wheels, portable grinders, threading machines, drill presses, portable drills, rolls, and shears; uses pipe wrenches, sledge hammers, punches, clamps, and other hand tools
15. specifications and knowledge of nuts and bolts
16. chemicals, lubricants and oils used in oiling and greasing equipment and components
17. importance of ensuring that tools and equipment are suitable, well maintained, calibrated and operating effectively
18. importance of following good housekeeping and fire prevention procedures
19. importance of following job instructions and defined maintenance procedures
20. components of generation system Components: Valves, pumps, compressors, motor, fans, blowers, gearbox etc.
21. importance of understanding abnormal sounds and vibrations in machines, coupling of pump and motor
22. importance of reporting problems in a timely manner
23. standard procedure on how to deal with electric shocks and rescue a person from electrocution to minimize harm



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- 24.** Personal Protective Equipment (PPE) and clothing that must be worn during the inspection, repair and maintenance activities and from where these can be obtained PPE: Safety helmet, safety glove, safety shoe, ear plugs, dust suppression masks, welding glasses during welding operations, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), safety rope, etc

Generic Skills (GS)

User/individual on the job needs to know how to:

- 1.** fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
- 2.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in English and/or local language
- 3.** convey and share technical information clearly using appropriate language
- 4.** clarify task related information with appropriate personnel or technical advise
- 5.** seek to improve and modify own work practices
- 6.** communicate and cooperate with others in the team for better results
- 7.** seek assistance from fellow team members
- 8.** make appropriate work-related judgments based on laid down policies and procedures
- 9.** plan and organize tasks to meet quantity and quality specifications
- 10.** identify impact of own work and actions on end customer
- 11.** identify changes in conditions, operations, and the environment to anticipate and address problems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out basic tasks related to mechanical maintenance and repair</i>	30	70	-	-
PC1. prepare and maintain the work area as per procedures or operational specification	3	7	-	-
PC2. identify the tools and equipment required for operation and maintenance of mechanical system in thermal power plant	3	6	-	-
PC3. obtain the tools and equipment identified following the organizational procedure	3	6	-	-
PC4. perform various activities to help the mechanic or engineer Activities: holding, handing over components, measuring, loading and unloading, etc. while assisting mechanics or mechanical operators in carrying out replacement of components such as radiators, hoses, fan belts, batteries, spark plugs, ignition cables, and other parts	3	7	-	-
PC5. assemble and disassemble valves, pumps, motors and other components to ensure proper functioning of the overall system	3	7	-	-
PC6. perform lubrication of various tools and related activities in accordance with the established procedures and charts Activities: Check and change oil, oil filters and air filters; fuel and service buses and other vehicles; replace hoses, belts, water pumps, batteries and similar equipment and parts	3	8	-	-
PC7. perform a regular condition check of various components and equipment Components and equipment: Fan belts, hoses, wiper blades and safety equipment; radiators and batteries	3	7	-	-
PC8. prepare and adjust various tools Prepare and adjust: Raise, lower, or position tools, equipment and materials using hoist or hand line	3	7	-	-
PC9. paint various tools related to electrical functions, as instructed	2	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. detect various signs of fault in the equipment Signs of fault: Vibrations, excessive sound, oil, air, gas, water and steam leakage or any similar abnormality	2	6	-	-
PC11. leave the work area in a safe and tidy condition on completion of the repair and maintenance activities	2	4	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1702
NOS Name	Perform basic tasks for repair and maintenance of equipment and components (Mechanical)
Sector	Power
Sub-Sector	Generation
Occupation	Operation & Maintenance; Erection, Installation and Commissioning Generation
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	11/02/2019
Next Review Date	10/02/2023
NSQC Clearance Date	22/08/2019

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- 1.Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack



Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1331.Apply basic health and safety practices for power related work	30	70	-	-	100	15
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N1701.Perform basic tasks for repair and maintenance of power generation system and components (Electrical)	30	70	-	-	100	40
PSS/N1702.Perform basic tasks for repair and maintenance of equipment and components (Mechanical)	30	70	-	-	100	40
Total	120	280	-	-	400	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
CPR	Cardiopulmonary Resuscitation
PPE	Personal Protective Equipment
PTW	Permit To Work
kVA	Kilo Volt Ampere
KW	Kilowatt
kWh	Kilowatt Hour
CO2	Carbon Dioxide

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.